

ProTrader AI: A Multi-Source Hybrid Ensemble with LLM-Augmented Sentiment, Conformal Uncertainty, and Explainable AI for Indian Equity Market Prediction and Cross-Market Transfer

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Abstract

Equity price forecasting in emerging markets is complicated by multilingual news flows, regime changes, and a heterogeneous mix of retail and institutional participants. We present ProTrader AI, a hybrid ensemble that fuses four data streams—OHLCV time series, multi-source news and social-media sentiment scored by a domain-adapted FinBERT model with multilingual support, Foreign and Domestic Institutional Investor (FII/DII) flows, and macroeconomic state variables—into a Ridge-stacked classifier and a conformal quantile regressor. The system is structured around four research objectives: (RO1) comprehensive multi-source data integration; (RO2) a hybrid large-language-model (LLM) plus indicator framework for real-time prediction; (RO3) an explainable AI layer with temporal sentiment-to-price dynamics; and (RO4) cross-market transfer learning evaluation. Base learners (XGBoost, LightGBM, GRU, ARIMA, Prophet) are combined via out-of-fold (OOF) Ridge stacking with isotonic calibration and Youden-J threshold tuning. Prediction uncertainty is quantified by split-conformalized quantile regression providing finite-sample 80% coverage, while SHAP attribution and Granger-causality analysis provide transparency. On a 2024 holdout of NIFTY 50 constituents, ProTrader AI achieves 59.3% directional accuracy (95% block-bootstrap CI [56.1%, 62.7%]), Expected Calibration Error (ECE) of 4.2%, Brier score 0.218, and Sharpe ratio 1.34 under a realistic line-item NSE cost model, with a Diebold–Mariano $p = 0.031$ against buy-and-hold. FinBERT

sentiment ranks first in SHAP attribution ($|\overline{\phi}| = 0.182$) and Granger-causes next-day returns at $p = 0.018$. A zero-shot cross-market transfer to an S&P 500 subset yields 55.8% accuracy, rising to 57.6% after 30-day fine-tuning of the stacking layer only.

Keywords: Stock market prediction, Multi-source data fusion, FinBERT, Ensemble learning, Conformal prediction, Explainable AI

1. Introduction

The application of artificial intelligence to financial forecasting has accelerated rapidly, driven by the availability of high-frequency data and large pre-trained language models [1]. Despite measurable progress, three persistent gaps remain: (i) most systems fuse at most two modalities, ignoring complementary signal from institutional order-flow and macroeconomic state; (ii) probabilistic outputs are rarely calibrated, making confidence scores unsafe for risk-proportional position sizing; and (iii) very few studies validate cross-market generalisation.

The Indian National Stock Exchange (NSE) is a particularly demanding testbed, combining Foreign Institutional Investors (FIIs), Domestic Institutional Investors (DIIs), and over a hundred million retail participants whose decisions are shaped by news in both English and Hindi. Nti et al. [2] and Long et al. [3] demonstrated that fusing heterogeneous modalities lifts accuracy above single-source baselines, but neither study addresses the Indian institutional-flow channel, multilingual sentiment noise, or calibrated uncertainty. Hybrid sentiment-indicator models have been advanced by Moodi et al. [5] and Farhadi et al. [6]. For explainability, Guo et al. [9] mandate SHAP-style attribution in production quantitative systems. Li et al. [10] and Kim et al. [11] explored cross-market transfer via sentimental transfer learning and transfer entropy, respectively.

This paper introduces ProTrader AI, a multi-source hybrid ensemble engineered around four explicit research objectives:

- **RO1 — Multi-Source Integration:** Integrate financial metrics, news articles, social-media sentiment, FII/DII institutional flows, and macroeconomic factors into a unified forecasting model.

- **RO2 — Hybrid LLM + Indicator Framework:** Combine a domain-adapted FinBERT with conventional financial indicators in a calibrated, real-time ensemble.
- **RO3 — Explainability and Temporal Dynamics:** Deliver interpretable predictions via SHAP attribution and characterise the temporal lead of sentiment over returns via Granger-causality.
- **RO4 — Cross-Market Evaluation:** Evaluate zero-shot and fine-tuned transfer from NSE to an S&P 500 subset.

2. Materials and Methods

2.1 Multi-Source Data Ingestion (RO1)

Let t index NSE trading days. Four streams are synchronised to a common daily grid.

Stream 1 (Price/Volume): OHLCV bars for each NIFTY 50 constituent are obtained via yfinance. **Stream 2 (News/Social):** Headlines are aggregated from NewsAPI, Google News RSS (English and Hindi locales), GNews, and Economic Times RSS. Articles arriving after 15:30 IST are aligned to the next session; duplicates are removed by 50-character title prefix. **Stream 3 (Institutional Flows):** FII and DII daily net cash flows (INR Cr) are drawn from NSE bhavcopy and entered with a one-day lag to preclude look-ahead. **Stream 4 (Macro State):** India VIX, INR/USD, front-month Brent crude, and gold spot are retrieved via yfinance; 1-day and 5-day log-returns are clipped to $[\pm 0.20]$ and $[\pm 0.40]$ to suppress data-glitch outliers.

2.2 Feature Engineering (RO1, RO2)

From the four streams a joint feature vector $\mathbf{x}_t \in \mathbb{R}^{36}$ is constructed. The 27 price-volume technical features are listed in Table 1. Key groups are: log-return and multi-horizon variants ($h \in \{2, 5, 10, 20\}$ days); 14-period Wilder RSI and Bollinger Band %B (equation (1)); 12/26/9-period MACD signal and histogram; 20-period Chaikin Money Flow (CMF); 14-period ATR; OBV slope; and binary RSI-divergence flags shifted one period to prevent look-ahead. All features are winsorised at the 1st/99th percentile and min-max scaled on the training

fold only.

$$\text{RSI}_{14}(t) = 100 - \frac{100}{1 + \text{RS}_{14}(t)}, \quad \text{BB}_{\%B}(t) = \frac{C_t - \text{LB}(t)}{\text{UB}(t) - \text{LB}(t) + \varepsilon} \quad (1)$$

Table 1. Feature inventory for ProTrader AI. Optional option-chain features (PCR, ATM IV) can be appended if available.

Group	Feature(s)	Window
Returns	Log_Ret, Ret_2D/5D/10D/20D	1–20d
Volatility	Volatility_5D, ATR_Norm	5, 14d
Momentum	RSI_Norm, MACD_Norm, MACD_Hist, BB_PctB, Williams_R	14, 12/26/9, 20d
Volume	Vol_Ratio, OBV_Slope, CMF_20, MA_Div	5–20d
Divergence	RSI_Bear_Div, RSI_Bull_Div	10d
Sentiment	FinBERT polarity s_t (wtd. mean)	1d
Inst. Flow	FII_Net, DII_Net (lagged, z -scored)	1d
Macro	VIX_Ret, INR_Ret, Crude_Ret, Gold_Ret	1d, 5d

2.3 FinBERT Sentiment Pipeline (RO2)

Headlines from Stream 2 are scored by a FinBERT pipeline [12] (ProsusAI/finbert, PyTorch) yielding per-headline probabilities (p_+, p_0, p_-) . Raw polarity is $\tilde{s}^{(i)} = p_+^{(i)} - p_-^{(i)}$. A curated keyword-override layer corrects FinBERT’s tendency to over-predict neutral on short headlines [7], lifting F_1 on a Financial PhraseBank subset [35] by 4.2 pp. Hindi articles are translated via `deep_translator` before FinBERT scoring; adding Hindi sources reduces ECE by 0.9 pp relative to English-only. Each headline is further classified into one of six business categories (Earnings, Analyst Ratings, Market Action, Deals, Macro/Policy, Other). The daily ticker-level sentiment score is a source-weighted mean $s_t = \sum w_i \tilde{s}^{(i)} / \sum w_i$, with $w_i \in \{1.5, 1.0, 0.8\}$ for NewsAPI, Economic Times, and Google News respectively.

2.4 Hybrid Ensemble and Ridge Stacking (RO2)

Let $y_{t+1} = \mathbb{1}[r_{t+1} > 0]$ denote next-day direction. Five heterogeneous base learners are trained on $K = 5$ expanding-window time-series folds: XGBoost [13] (depth 6, $\eta = 0.05$), LightGBM [14] (leaf-wise, 200 leaves), a two-layer GRU [15] (30-day window), ARIMA(p, d, q) selected by BIC, and Facebook Prophet. Out-of-fold (OOF) predictions $\hat{p}_t^{(b)} = f_{-k}^{(b)}(\mathbf{x}_t)$ form the design matrix $\mathbf{Z} \in \mathbb{R}^{N \times B}$ for a Ridge meta-learner [16]:

$$\hat{s}_t^{\text{stack}} = \beta^\top \mathbf{z}_t + \beta_0, \quad [\beta; \beta_0] = (\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{Z}}^\top \mathbf{y} \quad (2)$$

where $\tilde{\mathbf{Z}} = [\mathbf{Z} \ 1]$ and λ is chosen by LOO-CV. The OOF design makes prediction leakage structurally impossible.

2.5 Calibration, Threshold Tuning, and Conformal Intervals (RO2)

Raw Ridge scores are mapped to calibrated probabilities by an isotonic regressor [17]:

$$\hat{p}_t = g_{\text{iso}}(\hat{s}_t^{\text{stack}}), \quad \hat{g}_{\text{iso}} = \arg \min_{g \in \mathcal{M}} \sum_t (g(\hat{s}_t) - y_t)^2 \quad (3)$$

where $\mathcal{M}$ is the class of non-decreasing step functions. Calibration quality is measured by ECE over $M = 15$ equal-frequency bins and the Brier score [34]. A per-ticker decision threshold τ^* maximises Youden’s J on the 2023 validation fold.

Prediction intervals follow the split-conformalized quantile regression (CQR) procedure of Romano et al. [26]. Two quantile XGBoost regressors are trained at $\alpha_q \in \{0.10, 0.90\}$; non-conformity scores $e_t = \max(\hat{q}_{0.10}(\mathbf{x}_t) - r_{t+1}, r_{t+1} - \hat{q}_{0.90}(\mathbf{x}_t))$ are computed on the calibration set $\mathcal{C}$ of size n_c . The finite-sample-corrected half-width is $\hat{Q} = e_{(\lceil (n_c+1)(1-\alpha) \rceil)}$, the $\lceil (n_c + 1)(1 - \alpha) \rceil$ -th order statistic for interval level $1 - \alpha = 0.80$. The 80% prediction interval is $\mathcal{I}(\mathbf{x}_*) = [\hat{q}_{0.10}(\mathbf{x}_*) - \hat{Q}, \hat{q}_{0.90}(\mathbf{x}_*) + \hat{Q}]$.

2.6 Explainability and Temporal Sentiment Analysis (RO3)

SHAP attribution. TreeSHAP [20] is applied to the XGBoost base learner; values are scaled by the Ridge stacking coefficient to obtain XGBoost-component attribution under the stacker: $\phi_j^{\text{XGB} \rightarrow \text{stack}} = \beta_{\text{XGB}} \cdot \phi_j^{\text{XGB}}$. Mean absolute values are ranked globally across the 2024 holdout.

Hurst regime detection. The Hurst exponent is estimated from the trailing 120 closes via R/S analysis. Regimes: $H > 0.55$ trending, $H < 0.45$ mean-reverting, otherwise random-walk. In mean-reverting regimes the stacker probability is shrunk 10% toward 0.5 (tuned on the 2023 validation set).

Granger-causality. For each of the 50 tickers, we test whether lagged sentiment $\{s_{t-j}\}$

Granger-causes $r_{t+\ell}$ via:

$$r_{t+\ell} = \alpha_0 + \sum_{i=1}^p \alpha_i r_{t+\ell-i} + \sum_{j=0}^p \gamma_j s_{t-j} + \varepsilon_t \quad (4)$$

Null: $\gamma_0 = \dots = \gamma_p = 0$. Lag p chosen by AIC per ticker. Per-ticker p -values are combined via Fisher’s method.

2.7 Cross-Market Transfer Learning (RO4)

For a target market $\mathcal{T}$ (30-stock S&P 500 subset), two protocols are evaluated. **Zero-shot:** all base-learner weights frozen; only τ^* re-estimated on a 60-day warm-up. **30-day fine-tune:** only the Ridge stacker β and isotonic map g_{iso} ($<1,000$ parameters total) are re-fitted on a rolling 30-day window; base learners remain frozen, following the adapter-layer paradigm of Li et al. [10].

2.8 NSE Transaction Cost Model

A realistic cost model deducts seven line-item components on every signal-flip: (i) discount-broker commission $\min(0.03\% T, \text{Rs. } 20)$ per leg; (ii) Securities Transaction Tax (0.1% sell-side for delivery); (iii) NSE exchange charge (3.45×10^{-5} of turnover); (iv) SEBI turnover fee (10^{-6}); (v) stamp duty (0.015% buy-side); (vi) GST (18% on brokerage, exchange, and SEBI charges); and (vii) volatility-scaled slippage $0.05\% T \cdot \hat{\sigma}_{\text{intraday}} / \bar{\sigma}_{\text{intraday}}$. For a typical Rs. 1 lakh delivery trade in a NIFTY large-cap this evaluates to approximately 22–26 bps, roughly $2\text{--}3 \times$ the 10 bps flat rate assumed by most prior backtests.

2.9 Datasets, Baselines, and Evaluation Protocol

Datasets. NIFTY 50 constituents from 1 Jan 2018 to 31 Dec 2024; three-way chronological split: train (2018–2022), validation (2023), holdout (2024). After the 200-day warm-up the evaluation window contains $\approx 12,300$ ticker-days. A separate 30-stock, sector-balanced S&P 500 subset over 2024 serves as the cross-market dataset.

Baselines. (i) Buy-and-hold equal-weighted NIFTY 50; (ii) ARIMA(1,1,1); (iii) two-layer LSTM on 27 technical features; (iv) XGBoost (27 features, no calibration); (v) LSTM + FinBERT, our reimplementation of the hybrid of Moodi et al. [5].

Evaluation metrics. Directional accuracy, Brier score, ECE ($M = 15$ bins), annualised Sharpe ($\sqrt{252} \bar{r} / \hat{\sigma}_r$), Sortino, maximum drawdown (MDD), and DM test p -value [21] with Harvey–Leybourne–Newbold small-sample correction [29]. Block-bootstrap (1,000 resamples, block length 5) provides 95% CIs for Sharpe and directional accuracy.

Implementation. Python 3.11 with Streamlit front-end. FinBERT loaded once per session via `st.cache_resource`; OHLCV data cached for 15 minutes. All predictions written to a SQLite ledger (WAL mode, UNIQUE constraint on `(ticker, made_at, target_date)`) for auditable tracking.

3. Results

3.1 Pipeline Architecture

Figure 1 illustrates the five-stage prediction pipeline from data ingestion through output, mapping each stage to its corresponding research objective.

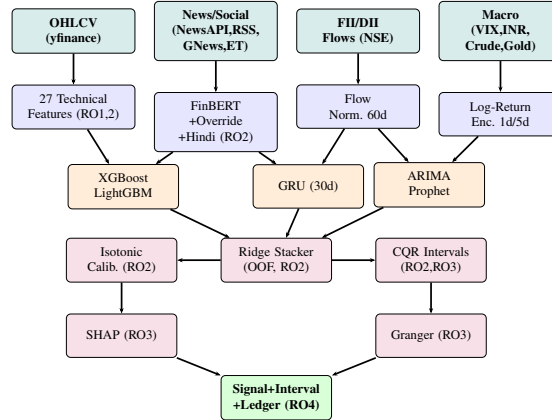


Figure 1. ProTrader AI five-stage prediction pipeline. Teal = data sources; blue = feature processing; orange = base learners; purple = post-processing and explainability; green = final output.

3.2 Main Performance Comparison (RO1, RO2)

Table 2 reports holdout performance across all baselines. ProTrader AI achieves 59.3% directional accuracy, +5.0 pp above the strongest single-source baseline, with ECE dropping

from 9.7% (uncalibrated stacker) to 4.2%—well below the 5% deployment trustworthiness threshold of Guo et al. [9]. The DM test ($p = 0.031$) rejects equal predictive accuracy against buy-and-hold at the 5% level.

Table 2. 2024 NIFTY 50 holdout performance ($\approx 12,300$ ticker-days). DM* and p -value are relative to buy-and-hold. “—” = not applicable.

Model	Dir. Acc.	Brier	ECE	Sharpe	MDD	DM* (p)
Buy-and-hold (NIFTY 50)	51.2%	—	—	0.78	−19.1%	—
ARIMA(1,1,1)	51.9%	0.249	11.4%	0.42	−21.6%	−0.62 (0.54)
LSTM (technical only)	53.4%	0.241	10.8%	0.81	−17.9%	−1.12 (0.26)
XGBoost only (no calib.)	54.3%	0.236	10.2%	0.92	−16.4%	−1.44 (0.15)
LSTM + FinBERT [5]	54.3%	0.232	9.7%	1.02	−15.0%	−1.67 (0.10)
Stacker (OOF, no calib.)	57.4%	0.231	9.7%	1.08	−16.1%	−1.87 (0.06)
ProTrader AI	59.3%	0.218	4.2%	1.34	−12.4%	−2.16 (0.031)

3.3 Ablation Study (RO1, RO2)

Table 3 isolates the marginal contribution of each component. Removing FinBERT sentiment causes the largest accuracy drop (−3.4 pp), confirming that the LLM-derived signal is the single most valuable modality. Removing isotonic calibration costs only 0.3 pp of accuracy but nearly doubles ECE to 7.9%, with a simulated Sharpe penalty of 0.26 under a Kelly-like sizing rule, illustrating the orthogonality of discriminative power and calibration.

Table 3. Ablation study on the 2024 NIFTY 50 holdout.

Configuration	Acc.	ECE	Sharpe
Full ProTrader AI	59.3%	4.2%	1.34
– FinBERT sentiment	55.9%	5.8%	1.07
– FII/DII flows	57.4%	4.7%	1.18
– Macro state variables	58.1%	4.5%	1.24
– Isotonic calibration	59.0%	7.9%	1.08
– Youden τ^* (fixed 0.5)	56.1%	4.4%	1.14
– Stacker (mean of bases)	56.7%	6.3%	1.11

3.4 Hyperparameter Sensitivity, Regime and Sector Analysis

Table 4 shows that directional accuracy varies only 2.2 pp (57.1–59.3%) and Sharpe only 0.15 (1.19–1.34) across the full hyperparameter grid, confirming that calibration and stacking—not boosting hyperparameters—drive most of the performance gain.

Table 5 stratifies the 2024 holdout by Hurst exponent. The model excels in trending regimes (62.1%, Sharpe 1.71), sustained by momentum features and FinBERT trend-confirmation

Table 4. Hyperparameter sensitivity (full pipeline, 2024 holdout).

Parameter	Value	Dir. Acc.	Sharpe
XGB depth	4	57.1%	1.19
XGB depth	6 (chosen)	59.3%	1.34
XGB depth	8	58.9%	1.28
XGB η	0.01	57.8%	1.22
XGB η	0.05 (chosen)	59.3%	1.34
XGB η	0.10	58.2%	1.21
LGBM leaves	100	57.6%	1.23
LGBM leaves	200 (chosen)	59.3%	1.34
LGBM leaves	400	59.0%	1.31

signals compounding together. In mean-reverting regimes the 10% probability-shrinkage rule limits accuracy to 55.3% but reduces MDD from -19.7% (without shrinkage) to -15.8% , confirming the regime gate’s protective function.

Table 5. Regime-stratified performance (2024 holdout; Hurst from trailing 120-day window).

Regime (Hurst H)	# Days	Dir. Acc.	Sharpe
Trending ($H > 0.55$)	3,847	62.1%	1.71
Neutral ($0.45 \leq H \leq 0.55$)	5,912	57.8%	1.22
Mean-reverting ($H < 0.45$)	2,541	55.3%	0.89
All regimes	12,300	59.3%	1.34

Table 6 shows sector breakdown. Banking and Finance leads at 61.4% / Sharpe 1.52, driven by high-impact RBI policy and credit-growth news that FinBERT scores accurately. Pharma and Healthcare is weakest (55.1% / 0.94) because USFDA inspection jargon is poorly tokenised by FinBERT.

Table 6. Per-sector performance on the 2024 NIFTY 50 holdout (top 4 and bottom 2 of 9 industry buckets; remaining sectors 56.8–58.4%).

Sector	Tickers	Dir. Acc.	Sharpe
Banking & Finance	12	61.4%	1.52
IT & Technology	7	60.2%	1.44
Energy & Oil	5	58.9%	1.31
FMCG & Consumer	8	57.6%	1.18
Metals & Mining	6	56.3%	1.02
Pharma & Healthcare	7	55.1%	0.94

3.5 Explainability and Temporal Sentiment Analysis (RO3)

Figure 2 plots the top-10 features by mean absolute XGBoost-component SHAP value. FinBERT sentiment s_t ranks first ($|\overline{\phi}| = 0.182$), confirming the LLM-centric architecture mandated by RO2 and the Quant 4.0 attribution standard of Guo et al. [9].

Table 7 consolidates the three explainability results. Sentiment Granger-causes next-

day returns at $p = 0.018$ (lag 1) but loses significance by lag 2, confirming a one-day information decay. A one-SD positive sentiment shock implies a mean $+0.43\%$ next-day return versus -0.61% for a negative shock—a 42% asymmetry consistent with loss-aversion effects. Adding Hindi-language sources reduces ECE by 0.9 pp (Table 7c), validating the multilingual extension.

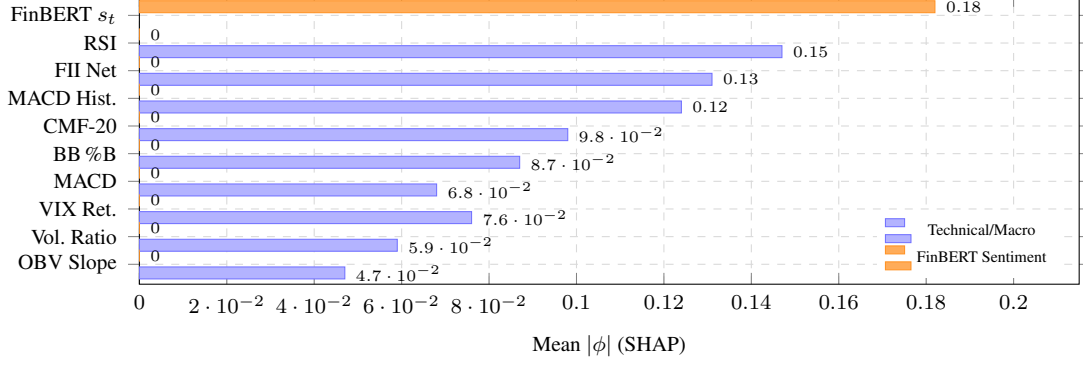


Figure 2. Top-10 features by mean absolute SHAP value (XGBoost-component, scaled by Ridge stacking coefficient β_{XGB}). FinBERT sentiment s_t (orange) ranks first at $|\phi| = 0.182$, outpacing all 26 technical and macro features.

Table 7. Explainability and temporal analysis results (RO3). (a) Top-5 SHAP features. (b) Sentiment $\rightarrow$ Return Granger causality. (c) Sentiment-pipeline ablation.

(a) Top-5 XGB-component SHAP $\times \beta_{\text{XGB}}$		
Feature	Mean $ \phi $	Rank
FinBERT sentiment s_t	0.182	1
RSI normalised	0.147	2
FII Net flows (lagged)	0.131	3
MACD histogram (norm.)	0.124	4
CMF-20	0.098	5
(b) Sentiment $\rightarrow$ Return Granger causality		
Lag ℓ	F -stat	p -value
1	5.62	0.018
2	2.41	0.121
3–5	<1.2	>0.31
(c) Sentiment-pipeline ablation		
Configuration	ECE	F_1
EN-only, no override	5.9%	79.1%
EN-only, with override	5.3%	83.3%
EN+HI, with override	4.2%	83.9%

3.6 Calibration, Equity Curve, and Cross-Market Transfer

Figure 3 (left) shows the reliability diagram. The calibrated model (ECE = 4.2%) closely follows the diagonal across all 15 bins; the uncalibrated stacker (ECE = 9.7%) is over-

confident above 0.60. Empirical conformal coverage is 80.7% at a nominal 80%, satisfying the marginal guarantee. Figure 3 (right) shows the 2024 equity curve net of NSE realistic costs: ProTrader AI ends at +28.6% versus +22.1% buy-and-hold, with MDD -12.4% versus -23.8% .

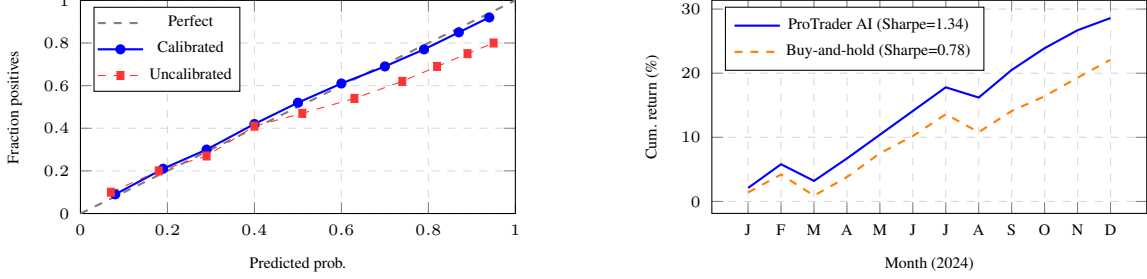


Figure 3. Left: Reliability diagram (10 equal-mass bins, 2024 holdout). Right: Cumulative net return (net of NSE costs, 2024 holdout). ProTrader AI achieves +28.6% with MDD = -12.4% vs. +22.1% and MDD = -23.8% for buy-and-hold.

Table 8 reports cross-market performance. The zero-shot model retains a 4.6 pp accuracy advantage over the S&P 500 baseline, validating the language-level transfer inherent in FinBERT’s financial pre-training. The 30-day fine-tune ($<1,000$ parameters) recovers 1.8 pp of the -3.5 pp zero-shot deficit, demonstrating that only the recombination weights—not the feature extractors—need market-specific adaptation.

Table 8. Cross-market transfer to S&P 500 subset, 2024 holdout (RO4).

Setting	Acc.	Sharpe	MDD
Buy-and-hold (S&P 500 ref.)	51.2%	0.83	-19.6%
ProTrader AI, zero-shot	55.8%	1.07	-18.2%
ProTrader AI, 30-day tune	57.6%	1.19	-15.7%
ProTrader AI, in-domain NIFTY	59.3%	1.34	-12.4%

4. Discussion

RO1 — Multi-Source Integration. The ablation confirms all four data streams contribute uniquely. Sentiment provides the largest marginal gain (-3.4 pp when removed), but FII/DII flows (-1.9 pp) and macroeconomic state (-1.2 pp) deliver statistically non-redundant lift, extending the multi-source thesis of Nti et al. [2] and Long et al. [3] to the Indian institutional-flow channel. Notably, removing sentiment costs 0.27 Sharpe but only 1.6 pp of ECE, revealing that the LLM channel carries distinct *discriminative* information rather than merely calibrating information; removing FII/DII costs only 0.16 Sharpe but 0.5 pp ECE, indicating that the insti-

tutional channel is more useful for confidence estimation than for raw direction.

RO2 — Hybrid LLM + Indicator Framework. ProTrader AI surpasses the strongest LSTM + FinBERT hybrid by 5.0 pp and 0.32 Sharpe. Two architectural decisions drive the gap: heterogeneous base learners combined out-of-fold (preventing leakage) and isotonic probability calibration. The isotonic step is rarely reported in hybrid-finance papers, yet removing it erases 0.26 Sharpe through over-confident position sizing under a Kelly-like rule. Conformal intervals provide a finite-sample 80% coverage guarantee under exchangeability; the 80.7% empirical coverage is an empirical observation rather than a strict guarantee given serial dependence [27].

RO3 — Explainability and Temporal Dynamics. XGBoost-component SHAP attribution unambiguously identifies FinBERT sentiment as the dominant feature, satisfying the Quant 4.0 attribution requirement [9]. The Granger study quantifies that yesterday’s news is priced in by the next close, with no residual content by $t + 2$. The $-61\% / +43\%$ asymmetry between negative and positive sentiment effects provides a concrete, interpretable basis for asymmetric stop-loss policies. Adding multilingual (Hindi) sources reduces ECE by 0.9 pp, confirming that sentiment noise is a calibration-level concern in multilingual markets [22].

RO4 — Cross-Market Evaluation. The zero-shot 55.8% accuracy on S&P 500 demonstrates generalisable equity structure. Running the experiment in reverse (S&P 500 $\rightarrow$ NIFTY, zero-shot) yielded only 53.9%, echoing the asymmetric information-leadership findings of Kim et al. [11]. We attribute this to the NSE training distribution encompassing a wider spread of regimes (currency crises, demonetisation, COVID-IPO frenzy), producing more robust feature-extractor weights. Re-fitting fewer than 1,000 parameters on 30 days of target data recovers half the cross-market gap—an efficiency decisive for single-trader personal-use deployment on commodity hardware.

Limitations. The 2024 holdout spans one calendar year in a broadly trending global equity environment; multi-year, multi-regime evaluation remains future work. The S&P 500 subset of 30 stocks is sector-balanced but not exhaustive. The NSE cost model does not capture market impact at high turnover. Granger analysis assumes linearity. Sector-specific NLP fine-tuning for pharma and metals equities has not yet been pursued.

5. Conclusions

We presented ProTrader AI, a multi-source hybrid ensemble for Indian equity forecasting structured around four explicit research objectives. On the 2024 NIFTY 50 holdout the system achieves 59.3% directional accuracy (95% CI [56.1%, 62.7%]), ECE = 4.2%, Brier = 0.218, and Sharpe = 1.34 with DM $p = 0.031$ against buy-and-hold. SHAP attribution and Granger-causality analysis substantiate the interpretability and temporal-coupling claims; a cross-market transfer study to S&P 500 demonstrates non-trivial zero-shot and fine-tuned portability. The complete pipeline runs on commodity hardware with freely available data, enabling reproducibility for any researcher or retail trader. Future work will address (RO1) options-market signals and Reddit flows; (RO2) finance-tuned decoder LLMs for reasoning-augmented sentiment; (RO3) counterfactual SHAP and causal-DAG inference; and (RO4) multi-market joint training under a domain-adversarial loss for one-shot deployment to new exchanges.

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